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CERTIFICATE

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This is to certify that

Yuvan Harshaj

bearing USN [1EP24CS181], Section [CSE-C], has successfully completed the Advanced UI/UX activity (Problem No. [64]) titled

"Unified RTL and Indic Design System"

& has submitted this report as required on 11-11-2026, during the academic year 2026-27.

Prasanth
Assistant Professor,
Dept. of CSE

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Abstract

The Unified RTL and Indic Design System aims to address the challenges of designing consistent, accessible and user-friendly interfaces for different writing systems and languages. Modern digital applications often face difficulties in supporting left-to-right (LTR), right-to-left (RTL) and complex Indic scripts due to variations in text direction, typography, font rendering, iconography, spacing and layout alignment. These challenges can affect readability, navigation and the overall user experience.

This project proposes a unified multilingual design system through a public transportation application named **TransitOne**. The system is designed to support English, Urdu, Kannada and Hindi while maintaining visual consistency and usability across different languages. The project follows a user-centred design approach involving user research, competitor analysis, persona development, empathy mapping, journey mapping and information architecture to identify user requirements and usability challenges.

Figma is used to create low-fidelity wireframes, reusable UI components, visual design elements and an interactive high-fidelity prototype. Antigravity IDE is used to develop a functional web demonstration using HTML, CSS and JavaScript, incorporating direction-aware layouts and multilingual typography. The prototype will undergo usability testing with real users and accessibility evaluation based on WCAG 2.2 AA guidelines. The project aims to demonstrate an adaptable and inclusive design system that improves multilingual readability, navigation, consistency and accessibility.

Furthermore, the proposed design system focuses on creating reusable and adaptable interface components that can automatically accommodate different writing directions and language-specific requirements. Special attention is given to appropriate icon mirroring, consistent spacing, responsive layouts, text expansion and proper rendering of complex characters. The system also considers accessibility requirements to ensure that users with different language preferences can interact with the application comfortably. By combining localisation principles with reusable design components, TransitOne demonstrates how a single design framework can support multiple languages without requiring completely separate interface designs. The project highlights the importance of inclusive design practices in developing modern digital applications and provides a foundation for building scalable multilingual user interfaces.

1. Problem Definition and Scope

Problem statement

In today's digital world, mobile applications are used by people belonging to different linguistic and cultural backgrounds. However, designing a single user interface that works effectively across multiple languages and writing systems remains a significant challenge. Most digital interfaces are primarily designed for left-to-right (LTR) languages such as English, making it difficult to provide a consistent experience for right-to-left (RTL) languages such as Urdu and complex Indic scripts such as Kannada and Hindi.

Different writing systems have unique requirements related to text direction, font rendering, character spacing, line height, alignment and overall layout structure. When an application is translated without considering these requirements, it can result in incorrectly aligned text, overlapping characters, inconsistent spacing, confusing navigation and improperly displayed icons. Additionally, automatically mirroring every interface element may create usability issues because certain icons, symbols and visual elements should retain their original orientation.

The assigned problem focuses on developing a **Unified RTL and Indic Design System** that provides a common design framework capable of supporting multiple writing systems while maintaining consistency, readability and accessibility. The system must intelligently adapt layouts, typography, icons and interface components according to the selected language and its writing direction.

To demonstrate this approach, the project proposes **TransitOne**, a multilingual public transportation application supporting English, Urdu, Kannada and Hindi. The application will demonstrate how a single reusable design system can accommodate different linguistic requirements without creating completely separate interfaces for each language.

The primary objective is to develop an inclusive, adaptable and scalable interface that provides users with a consistent and comfortable experience regardless of their language preference or writing direction.

Stakeholders and target users

The Unified RTL and Indic Design System involves different stakeholders who contribute to the development, implementation and usage of multilingual digital interfaces. The primary focus is to ensure that users from different linguistic and cultural backgrounds can interact with

the application comfortably and efficiently.

1. Primary Stakeholders

- **End Users:** People who use mobile applications in English, Urdu, Kannada and Hindi and require an interface that is easy to understand and navigate.
- **UI/UX Designers:** Professionals responsible for designing layouts, typography, icons, reusable components and consistent visual experiences across different writing systems.
- **Frontend Developers:** Developers responsible for implementing direction-aware layouts, language switching, font rendering and responsive interface components.
- **Accessibility Specialists:** Individuals who evaluate the interface to ensure readability, accessibility and usability for people with different needs.
- **Transport Service Providers:** Organisations that provide public transportation information and services through digital platforms.

2. Target Users

The proposed TransitOne application primarily targets the following user groups:

- **English-speaking users:** Users who prefer English interfaces and left-to-right navigation.
- **Kannada-speaking users:** Users who prefer accessing transport information in their regional language with properly rendered Kannada characters.
- **Hindi-speaking users:** Users who require readable Devanagari text and familiar navigation elements.
- **Urdu-speaking users:** Users who require right-to-left layouts, correctly aligned text and appropriate directional navigation.
- **Multilingual users:** Individuals who frequently switch between different languages and expect a consistent interface across language settings.

3. User Requirements

The target users require an application that provides easy language switching, readable typography, properly aligned content, understandable icons, consistent navigation and accessible interface components. The design system aims to fulfil these requirements through reusable and adaptable design elements that support different writing directions and language-specific characteristics.

By considering the needs of these stakeholders and target users, TransitOne aims to demonstrate an inclusive multilingual interface that maintains usability and visual consistency across different scripts and languages.

Goals and success metrics

The primary goal of the Unified RTL and Indic Design System is to develop a consistent, adaptable and accessible interface that supports multiple languages and writing directions within a single design framework. The proposed **TransitOne** application will demonstrate how a unified design approach can improve the usability of multilingual digital applications.

1. Project Goals

- To develop a unified design system that supports Left-to-Right (LTR), Right-to-Left (RTL) and complex Indic scripts.
- To provide multilingual support for English, Urdu, Kannada and Hindi.
- To implement appropriate layout mirroring for directional interface elements without affecting non-directional icons.
- To ensure proper font rendering, character spacing, text alignment and line height for different writing systems.
- To create reusable UI components that maintain visual consistency across all supported languages.
- To provide a simple and convenient language-switching mechanism.
- To develop responsive layouts that work across different screen sizes and devices.
- To improve accessibility through readable typography, sufficient colour contrast and clear navigation.
- To validate the design through user research and usability testing.
- To demonstrate the proposed design system through an interactive Figma prototype and a functional web implementation.

The design system will focus on maintaining a seamless user experience while adapting to the unique requirements of different writing systems. It will minimize inconsistencies in layouts and ensure that translated content is displayed correctly across all supported languages. Special attention will be given to text expansion, font compatibility and directional navigation.

The system will also encourage component reusability, reducing the need to create separate interfaces for individual languages. User feedback and usability evaluation will help identify design limitations and guide necessary improvements. Ultimately, the project aims to establish a scalable foundation for developing inclusive and multilingual digital applications.

2. Success Metrics

The success of the project will be evaluated using the following measurable criteria:

Metric	Success Criteria
Language support	Successful implementation of all four selected languages
Layout direction	Correct LTR and RTL layout behaviour
Typography	No overlapping, clipping or incorrectly rendered characters in tested content
Icon consistency	Correct mirroring of directional icons
Language switching	Users can change languages without losing navigation context
Reusability	Common components maintain consistent styling across language modes
Responsive design	Interface remains usable on mobile and desktop screen sizes
Usability testing	At least 5 users participate in prototype evaluation
Task completion	Target at least 80% successful completion of defined usability tasks
Accessibility	Evaluation against applicable WCAG 2.2 AA requirements

3. Expected Outcomes

The expected outcome is a unified multilingual design system that demonstrates consistent layouts, script-specific typography, appropriate iconography and accessible navigation across different writing systems. The final solution will be evaluated using participant feedback, usability testing observations and accessibility checks to identify areas for further improvement.

Ultimately, the project aims to demonstrate that a single adaptable design framework can

support multiple writing systems without compromising usability, accessibility or visual consistency. It will serve as a foundation for developing inclusive, scalable and localisation-friendly digital applications that can accommodate users from diverse linguistic backgrounds.

Constraints and assumptions

The development of the Unified RTL and Indic Design System involves certain technical, design-related and practical constraints that must be considered to ensure successful implementation. Since the project focuses on supporting multiple writing systems within a single interface, special attention must be given to language compatibility, layout behaviour and accessibility.

1. Constraints

- **Language Compatibility:** Supporting multiple writing systems requires appropriate fonts and rendering support to display characters correctly without missing glyphs or overlapping text.
- **Layout Adaptability:** The interface must accommodate both left-to-right and right-to-left layouts while maintaining consistent spacing, alignment and navigation.
- **Typography Limitations:** Different scripts have different character structures, font metrics and line-height requirements, making it challenging to maintain a uniform typography system.
- **Icon Direction:** Certain directional icons require mirroring, whereas universal symbols and non-directional icons must retain their original orientation.
- **Screen Responsiveness:** Interface components must adapt to different screen sizes without affecting readability or usability.
- **Development Time:** The project must be completed within the available academic timeline, limiting the number of features and advanced functionalities that can be implemented.
- **Tool Limitations:** The implementation depends on the capabilities of Figma, Antigravity IDE, available fonts and standard web technologies.
- **User Research Availability:** The quality of usability evaluation depends on the availability of participants representing different language preferences.

2. Assumptions

- Users will be able to select their preferred language through a dedicated language-switching option.
- Appropriate font families will be available for English, Urdu, Kannada and Hindi.
- Users will have access to devices and browsers that support the selected languages and character sets.
- The public transportation application will use sample route information and demonstration content rather than a live transportation database.
- The same design framework will be capable of adapting to different writing directions through reusable components and direction-aware layout rules.
- Users will be familiar with basic mobile application interactions such as searching, selecting options and navigating between screens.
- The prototype will primarily demonstrate the visual and interactive aspects of multilingual localisation rather than complete commercial application functionality.

These constraints and assumptions will help establish realistic project boundaries while ensuring that the main objectives of multilingual support, usability and accessibility remain the primary focus.

Scope: what the design will and will not cover

The scope of the Unified RTL and Indic Design System is to develop a unified, adaptable and reusable design framework that supports multiple languages and writing directions while maintaining consistency, readability and accessibility. The proposed TransitOne public transportation application will serve as the demonstration platform for implementing the design system. The project will focus on supporting English, Urdu, Kannada and Hindi through a common interface that adapts to the requirements of different writing systems.

The design will cover multilingual interface development, including Left-to-Right (LTR) and Right-to-Left (RTL) layouts, appropriate typography, font rendering, text alignment, spacing and icon adaptation. It will also include reusable UI components such as buttons, search fields, navigation bars, cards and language selectors. Special attention will be given to responsive layouts, language switching, directional icon mirroring and accessibility considerations. Figma will be used to create wireframes, visual designs and interactive prototypes, while Antigravity IDE will support the development of a functional web demonstration using

HTML, CSS and JavaScript.

Overall, the project will concentrate on demonstrating how a single design system can accommodate different writing systems without compromising usability, visual consistency or accessibility. The final outcome will include a reusable design framework, an interactive Figma prototype and a functional multilingual web demonstration.

2. User Research

Research method and questions asked

The research was planned using a qualitative approach to understand users' language preferences and their experiences with multilingual digital applications. Semi-structured interviews were selected to collect information about difficulties related to text alignment, typography, icon representation, navigation and language switching.

Five or more participants with different language backgrounds were considered to understand the requirements of users interacting with different writing systems. The research focused on English, Hindi, Kannada, Urdu and other regional language experiences to identify common usability challenges.

The following questions were prepared for the interviews:

1. Which language do you primarily use on your smartphone?
2. Do you prefer using applications in English or your native language?
3. Have you experienced problems with text alignment or font rendering?
4. Have you used applications that support right-to-left languages?
5. Do you find navigation icons confusing when the language changes?
6. Do you prefer icons with text labels or icons alone?
7. Have you experienced overlapping or incorrectly displayed text?
8. What difficulties do you face while using multilingual applications?
9. Would you prefer an application that allows instant language switching?
10. What improvements would make multilingual applications easier to use?

The collected responses will be analysed to identify common usability problems and translate them into practical design requirements for the proposed TransitOne application.

Participant details

The user research involves participants from different linguistic backgrounds to understand their experiences, preferences and expectations while using multilingual mobile applications. Participants were selected based on their familiarity with different languages and their experience interacting with digital interfaces.

The study focuses on identifying challenges related to language selection, text readability, typography, navigation, icon representation and interface alignment. The selected participants provide different language perspectives that help in understanding the requirements of a unified multilingual design system.

The following table presents the participant details considered for the research:

Sl. No.	Participant ID	Participant Name	Language Preference
1	P1	V. Madhusudhan	English and Hindi
2	P2	Shreyas M.	English and Kannada
3	P3	Sivanesan S.	Tamil and English

The participants represent varied language preferences and provide an opportunity to examine how different writing systems influence the usability of mobile applications. Their responses and observations will be used to identify common design challenges and establish requirements for the proposed TransitOne application.

The collected information will also support the development of user personas, empathy maps and journey maps in the subsequent stages of the project.

Key findings and direct observations

The user research focuses on understanding the challenges faced by individuals while using mobile applications across different languages. The selected participants represent different linguistic backgrounds, including English, Hindi, Kannada and Tamil. Their language preferences provide an opportunity to understand the importance of multilingual support and localisation in digital interfaces.

The following key areas have been identified for investigation:

1. Language Preferences: Users may have different language preferences depending on their familiarity, comfort and the type of application they are using.

2. Typography and Readability: Different writing systems require suitable fonts, appropriate font sizes and sufficient line spacing to ensure clear text presentation.

3. Layout and Alignment: Interfaces must adapt their alignment and reading direction according to the selected language without affecting the overall design structure.

4. Icon Representation: Directional icons may require mirroring in RTL interfaces, while universal symbols should maintain their original orientation.

5. Language Switching: A simple and accessible language selector is important for users who frequently interact with applications in multiple languages.

6. Mixed-Language Content: Combining English text with regional scripts can create alignment and rendering challenges that must be considered during interface development.

Overall Observation:

The research highlights the importance of developing multilingual interfaces that consider the unique requirements of different writing systems rather than simply translating existing content. These observations will guide the design of TransitOne by focusing on readable typography, adaptable layouts, consistent iconography and convenient language switching.

Competitor or existing solution analysis

A comparative analysis of existing mobile applications was conducted to understand how different platforms approach multilingual interface design, navigation and accessibility. Three applications, Google Pay, WhatsApp and IRCTC Rail Connect, were selected because they represent different categories of digital services and provide useful references for the proposed TransitOne application.

1. Google Pay

Google Pay provides support for multiple Indian languages, including Hindi, Kannada and Tamil. It offers language preferences that help users interact with payment-related services in familiar languages. Its multilingual approach provides a useful reference for language selection and localisation. However, our project will further explore how typography, text alignment and reusable components can be managed consistently across different writing systems.

2. WhatsApp

WhatsApp provides a multilingual communication environment where users interact with

different scripts and languages. It offers useful examples of mixed-language content, message alignment and text presentation. Its interface provides a reference for understanding right-to-left text handling and the challenges of displaying multiple writing systems within the same conversation.

3. IRCTC Rail Connect

IRCTC Rail Connect is a public transportation application that provides train search, journey details, ticket booking and passenger information. It serves as a useful reference for understanding transport-related navigation, information organisation and booking interfaces. Its features provide inspiration for the structure and functionality of the proposed TransitOne application.

Overall Comparative Observation

The analysis of these applications provides useful references for multilingual support, navigation and information presentation. TransitOne aims to combine these design considerations within a single reusable framework, with additional emphasis on LTR and RTL adaptation, Indic typography, selective icon mirroring and consistent interface components.

Insights that will drive the design

The following insights will drive the design of the proposed system:

1. **Simple Language Switching:** Provide a clearly visible language selector so users can easily switch between English, Urdu, Kannada and Hindi.
2. **Direction-Aware Layout:** Support both left-to-right (LTR) and right-to-left (RTL) layouts without disturbing the overall interface structure.
3. **Suitable Typography:** Use appropriate fonts for each script to improve readability and ensure that characters are displayed correctly.
4. **Consistent Spacing:** Maintain proper spacing, alignment and line height to prevent text overlapping or clipping, especially in Indic scripts.
5. **Selective Icon Mirroring:** Mirror only directional icons, such as back and forward arrows, while keeping non-directional icons unchanged.
6. **Reusable Components:** Develop reusable buttons, navigation bars, search fields and information cards that adapt to different languages and directions.
7. **Accessibility and Responsiveness:** Ensure that the interface remains readable,

accessible and usable across different screen sizes and devices.

These insights will be applied during wireframing, visual design, prototype development and usability testing.

No.	Participant profile	Method	Expected observation
1	V. Madhusudhan – English and Hindi	Semi-structured interview	Prefers simple language switching and clear text presentation. Consistent font size and readable Hindi text are important.
2	Shreyas M. – English and Kannada	Semi-structured interview	Needs proper Kannada font rendering, sufficient spacing and clearly aligned interface elements.
3	Sivanesan S. – Tamil and English	Semi-structured interview	Requires readable Tamil text and consistent navigation when switching between English and Tamil.

3. Personas and Empathy Map

Persona 1: Bilingual Commuter

Name: V. Madhusudhan

Language Preference: English and Hindi

User Type: Public Transportation App User

Profile:

V. Madhusudhan is a bilingual smartphone user who communicates in both English and Hindi. He represents users who prefer simple navigation, readable text and the flexibility to use an application in their preferred language. While using TransitOne, he needs a consistent interface that maintains its structure when switching between English and Hindi.

Goals:

- Easily switch between English and Hindi.
- View route and travel information clearly.
- Navigate the application without confusion.
- Read text comfortably in both languages.

Needs:

- A simple and visible language selector.
- Clear and readable Hindi typography.
- Consistent spacing and alignment.

- Familiar icons with clear labels.

Pain Points:

- Poorly displayed or difficult-to-read text.
- Confusing navigation after changing languages.
- Inconsistent alignment between language versions.
- Unclear icons without supporting labels.

Design Opportunity:

TransitOne should provide a simple language-switching option, readable typography and reusable interface components that maintain consistency across English and Hindi.

Research Connection: This persona is based on Participant 1's English and Hindi language profile and the design considerations identified in Section 2. The goals and pain points are provisional and should be validated through actual user feedback.

Persona 2: Bilingual Commuter

Name: Shreyas M.
Language Preference: English and Kannada
User Type: Public Transportation App User

Profile:

Shreyas M. is a bilingual smartphone user who uses both English and Kannada. He represents users who expect mobile applications to display regional language content clearly without affecting the overall layout. While using TransitOne, he needs readable Kannada text, proper spacing and consistent navigation across both languages.

Goals:

- Use the application comfortably in English and Kannada.
- Search for routes and view travel information easily.
- Navigate through a simple and consistent interface.
- Access information without text display issues.

Needs:

- Proper Kannada font rendering.
- Suitable text size and line spacing.

- Clear navigation buttons and icons.
- Consistent layout when switching languages.

Pain Points:

- Incorrect Kannada character rendering.
- Text overlapping or insufficient spacing.
- Misaligned interface elements.
- Difficulty understanding icons without labels.

Design Opportunity:

TransitOne should use suitable Kannada typography, flexible layouts and reusable UI components to maintain readability and consistency when switching between English and Kannada.

Research Connection: This persona is based on Participant 2's English and Kannada language profile. The goals and pain points are provisional design considerations and should be validated through actual user feedback.

Empathy Map: Says, Thinks, Does and Feels

1. SAYS

- "I want to use the application in my preferred language."
- "The text should be easy to read."
- "Navigation should be simple and clear."
- "I should be able to change languages easily."

2. THINKS

- Can I understand all the information in my preferred language?
- Will the layout remain consistent after changing languages?
- Are the fonts suitable for Kannada, Hindi and other scripts?
- Can I identify navigation icons easily?

3. DOES

- Switches between English and regional languages.
- Searches for routes and travel information.

- Uses navigation buttons and icons.
- Reads route details before planning a journey.

4. FEELS

- Comfortable when information is displayed in a familiar language.
- Confident when navigation is simple and consistent.
- Confused when text or icons are unclear.
- Frustrated when text overlaps or is displayed incorrectly.

Design Implication:

The empathy map highlights the importance of readable multilingual typography, simple language switching, direction-aware layouts and familiar navigation elements. These considerations will guide the design of the TransitOne application.

Pain Points and Unmet Needs

Users who interact with multilingual applications may experience several difficulties when switching between different languages and writing systems. These challenges can affect readability, navigation and the overall user experience. In the proposed TransitOne application, special attention is given to the following pain points and unmet needs.

Pain Points:

1. **Incorrect Text Rendering:** Kannada, Hindi and Urdu characters may not display properly if suitable fonts and rendering support are not provided.
2. **Layout Inconsistency:** Interface elements may become misaligned when switching between left-to-right and right-to-left languages.
3. **Navigation Confusion:** Directional icons such as back and forward arrows may become confusing if they are not mirrored correctly.
4. **Insufficient Spacing:** Different scripts require appropriate line height, spacing and text alignment to maintain readability.
5. **Difficult Language Switching:** Users may find it inconvenient when language selection is difficult to locate or use.
6. **Mixed-Script Issues:** Combining English words with regional language text can create alignment and display problems.

Unmet Needs:

Users need a unified interface that supports multiple languages without changing the overall design structure. They require suitable fonts, readable text, consistent navigation, flexible layouts and a simple language-switching option. They also need an accessible and responsive interface that works properly across different devices.

Addressing these needs will help TransitOne provide a more inclusive, consistent and user-friendly multilingual experience.

Problem reframed as a design question

How might we design a unified and accessible design system that supports LTR, RTL and complex Indic scripts? How can we maintain consistent typography, spacing, icons and navigation across different languages? How can we make the interface readable, responsive and user-friendly for everyone, regardless of their language preference?

4. User Journey and Information Architecture

End-to-end user journey with emotional highs and lows

The user journey explains how a commuter interacts with the **TransitOne** application, from opening the app to successfully finding travel information.

Stage	User Action	User Experience	Emotion
1. Open App	Opens TransitOne	Views the welcome screen	Curious
2. Select Language	Chooses English, Urdu, Kannada or Hindi	Interface adapts to the selected language	Happy
3. Home Screen	Explores available options	Understands the navigation and features	Comfortable
4. Search Route	Enters starting point and destination	Searches for available routes	Focused
5. View Results	Checks route details and travel information	Compares available options	Confident
6. Select Route	Chooses a suitable route	Views journey details	Satisfied
7. Complete Task	Confirms the selected journey information	Successfully completes the task	Happy

Emotional Highs and Lows: Users may feel confused when text is difficult to read or navigation changes unexpectedly after switching languages. They feel more comfortable and confident when the interface displays readable text, familiar icons and consistent layouts.

Task flows for the three most critical tasks

Task Flow 1: Changing the Application Language

Open TransitOne → Navigate to Settings → Select Language → Choose preferred language → Interface updates automatically → Return to Home Screen.

Task Flow 2: Searching for a Route

Open Home Screen → Select Route Search → Enter Starting Location → Enter Destination → Tap Search → View Available Routes → Select a Route → View Journey Details.

Task Flow 3: Viewing Journey Details

Open TransitOne → Search for a Route → Select the Required Route → View Route Information → Check Stops and Travel Details → Return to Home Screen or Search Again.

These three task flows focus on the most important activities and demonstrate how users can interact with TransitOne through a simple and consistent navigation structure.

Sitemap or navigation structure

TransitOne

- Welcome Screen
    - Language Selection
- Home Screen
    - Route Search
    - Recent Searches
    - Favourite Routes
- Route Details
    - Stops and Travel Information
- Settings
    - Language Preferences
    - Accessibility Options

Content and feature Prioritisation

The features of TransitOne are prioritised based on their importance and usefulness to users. High-priority features include language selection, LTR and RTL layout support, readable typography, route search and clear navigation icons. Medium-priority features include favourite routes, recent searches, responsive layouts and accessibility settings. Low-priority features include application information and additional visual customisation. This prioritisation ensures that the essential features are developed first, providing users with a simple, consistent and accessible multilingual experience across different languages and devices.

5. Ideation and Concept Selection

Rough sketches or rapid ideas (include scanned pages)

IInitial ideas were explored through rough sketches and simple layout arrangements. Different screen layouts were considered for language selection, home navigation, route search and settings. Special attention was given to text alignment, icon placement, typography and layout adaptation for LTR, RTL and Indic scripts. These initial ideas helped identify a suitable direction for the TransitOne application.

At least three distinct concepts

Concept A – Basic Multilingual Interface

A simple application layout with a language selector and translated text for different languages.

Concept B – Direction-Aware Interface

A flexible interface that automatically adapts alignment, navigation and directional icons according to the selected language.

Concept C – Unified Multilingual Design System

A reusable design system supporting English, Urdu, Kannada and Hindi with adaptable typography, icons, spacing and responsive layouts.

Evaluation of concepts against criteria

The three concepts were evaluated based on multilingual support, readability, layout consistency, accessibility and ease of implementation. Concept A provides basic language support but has limited layout adaptation. Concept B handles text direction effectively but

provides fewer reusable design features. Concept C offers a more complete solution by combining multilingual support, direction-aware layouts and reusable components.

Concept C also provides better flexibility for future improvements and additional language support. It maintains a consistent user experience across different devices and writing systems. It also reduces the need to create separate layouts for each language. Therefore, Concept C is considered the most suitable option for achieving the project's objectives.

Chosen Concept: Concept C – Unified Multilingual Design System

After evaluating all three concepts, Concept C was selected as the final design direction for the TransitOne application. It provides a complete solution for supporting English, Urdu, Kannada and Hindi within a single unified interface.

Concept C was selected because it combines multilingual support, direction-aware layouts, suitable typography, selective icon mirroring and reusable UI components. Unlike the basic multilingual approach, it adapts the interface according to the writing direction and language-specific requirements.

The selected concept also focuses on maintaining consistent spacing, alignment and readability across different screen sizes. It reduces the need to create separate interface designs for individual languages and makes the system easier to maintain and improve.

The following criteria were considered while selecting the final concept:

- **Multilingual Support:** Ability to support multiple languages and writing systems.
- **Layout Adaptability:** Proper handling of LTR and RTL directions.
- **Typography and Readability:** Suitable fonts, spacing and text alignment.
- **Consistency:** Maintaining a uniform appearance across all language versions.
- **Accessibility:** Providing a simple and user-friendly experience.
- **Reusability:** Supporting reusable components for future development.
- **Implementation Feasibility:** Ability to develop the design using Figma, HTML, CSS and JavaScript.

Concept C satisfies these requirements more effectively and provides a strong foundation for the proposed unified design system.

Final Decision: Concept C was selected for implementation because it offers the best combination of flexibility, accessibility, consistency and multilingual adaptability.

Criteria	Concept A	Concept B	Concept C
Multilingual support	Good	Good	Excellent
LTR/RTL adaptation	Limited	Excellent	Excellent
Typography	Basic	Good	Excellent
Layout consistency	Good	Good	Excellent
Accessibility	Basic	Good	Excellent

6. Low-Fidelity Wireframes

Wireframes of key screens

The low-fidelity wireframes of TransitOne were created to define the basic structure, layout and navigation of the application before applying colours and visual styling. Simple boxes, lines and placeholders were used to represent the interface elements.

The following key screens were considered:

1. **Welcome and Language Selection Screen:** Allows users to select English, Urdu, Kannada or Hindi.
2. **Home Screen:** Displays the application name, route search option, recent searches and navigation menu.
3. **Route Search Screen:** Provides input fields for starting location and destination.
4. **Route Results Screen:** Displays available routes and basic travel information.
5. **Route Details Screen:** Shows selected route information, stops and journey details.
6. **Settings Screen:** Provides language preferences and accessibility options.

These wireframes focus on creating a simple and consistent layout that can adapt to different writing directions and languages.

Annotations Explaining Layout and Interaction Decisions

The low-fidelity wireframes were designed to maintain a simple structure, clear navigation and easy interaction.

1. **Language Selection:** Positioned prominently on the welcome screen to help users select their preferred language easily.
2. **Navigation Bar:** Provides quick access to important sections such as Home, Search and Settings.

3. **Search Fields:** Starting location and destination fields are arranged clearly to simplify route searching.
4. **Action Buttons:** Important buttons are placed in easily accessible positions for better interaction.
5. **Flexible Layout:** The interface is designed to adapt to LTR and RTL writing directions without disturbing the overall structure.
6. **Typography Space:** Sufficient space is provided for English, Urdu, Kannada and Hindi text to prevent overlapping.
7. **Icon Placement:** Directional icons are designed to mirror according to the selected language, while non-directional icons remain unchanged.
8. **Consistent Components:** Similar buttons, input fields and cards follow a common layout across different screens.

These layout and interaction decisions help create a consistent, accessible and user-friendly experience across all supported languages.

Quick feedback from at least two people

The initial wireframes were reviewed to identify possible improvements in layout, navigation and usability.

Feedback 1:

1. The language selection option should be more visible.
2. Language names should be clearly displayed.
3. Navigation buttons should have understandable labels.
4. The layout should remain consistent after changing languages.

Feedback 2:

5. Search fields should have sufficient spacing.
6. Starting location and destination should be clearly identified.
7. Important buttons should be easy to locate.
8. Text should remain readable in Kannada, Hindi and Urdu.

Changes made after feedback

Based on the feedback received, the following improvements were considered in the wireframe design:

1. Improved the visibility of the language selection option.
2. Added clear labels for supported languages.
3. Refined the navigation structure for easier access.
4. Maintained consistent positioning of interface elements.
5. Increased spacing between route search fields.
6. Improved the placement of important action buttons.
7. Provided sufficient space for different script characters.
8. Planned direction-aware layouts for both LTR and RTL languages.

7. Visual Design and Design System

Colour palette with contrast ratios

The TransitOne application uses a clean and modern colour palette to provide a professional, comfortable and accessible user experience.

Colour	Hex Code	Usage
Primary Navy Blue	#123B5D	Headers, navigation and primary buttons
Secondary Teal Blue	#1D6F8A	Secondary buttons and highlights
Accent Yellow	#F4B942	Important highlights and indicators
Background White	#FFFFFF	Main background
Light Background	#F7F9FC	Cards and content sections
Text Dark	#1F2937	Main text and headings
Muted Grey	#5B6573	Secondary text
Brand Orange	#FF8A00	Primary CTA buttons, active state chips, and brand identity
Peach / Amber Tint	#FFF0E6	Active navigation pill and soft highlighted containers

Contrast Ratios:

- Navy Blue on White: 11.61:1
- Teal Blue on White: 5.69:1
- Dark Text on White: 14.68:1
- Muted Grey on White: 5.91:1

- Dark Text on Yellow: 8.29:1

The selected text and background combinations meet the WCAG AA contrast requirement for normal-sized text. These ratios are based on the listed colour combinations.

Typography and Type Scale

Typography is an important part of the TransitOne design system because different languages require suitable font styles and character spacing. The system uses script-specific fonts to ensure proper text rendering and readability across all supported languages.

Font Selection:

- English: Noto Sans
- Kannada: Noto Sans Kannada
- Hindi: Noto Sans Devanagari
- Urdu: Noto Naskh Arabic

Typography Scale:

Text Element	**Font Size**	**Font Weight**
Main Heading	28 px	Bold
Section Heading	22 px	Semibold
Subheading	18 px	Medium
Body Text	16 px	Regular
Button Text	14–16 px	Semibold
Caption	12 px	Regular

The design maintains appropriate line height, text spacing and alignment for each script. Urdu text follows right-to-left alignment, while English, Kannada and Hindi follow left-to-right

alignment. Flexible text containers are used to prevent overlapping and clipping. This typography system provides a consistent and readable experience across all four languages.

Spacing, Grid and Layout Rules

TransitOne follows a consistent spacing and layout system to maintain visual balance, readability and usability across different languages and screen sizes.

Spacing Rules:

- Uses an 8-point spacing system (8px, 16px, 24px and 32px).
- Maintains sufficient spacing between text, buttons and input fields.
- Provides adequate padding inside cards and interface components.
- Uses consistent margins to maintain alignment.

Grid and Layout Rules:

- Uses a flexible grid system to organise interface elements.
- Maintains consistent alignment across all application screens.
- Supports responsive layouts for mobile, tablet and desktop devices.
- Uses flexible containers to accommodate different text lengths.
- Applies LTR and RTL layouts according to the selected language.
- Uses CSS logical properties such as margin-inline and padding-inline for direction-aware spacing.

These rules help prevent overlapping, maintain proper alignment and provide a consistent user experience across English, Urdu, Kannada and Hindi.

Reusable Components and Their States

TransitOne uses reusable UI components to maintain consistency and simplify the design process across different screens and languages.

Main Components:

1. **Buttons:** Primary and secondary buttons for user actions.
2. **Input Fields:** Fields for entering starting locations and destinations.
3. **Navigation Bar:** Provides access to Home, Search and Settings.
4. **Language Selector:** Allows users to switch between English, Urdu, Kannada and Hindi.

5. **Information Cards:** Displays route details and travel information.
6. **Toggle Controls:** Used for language and accessibility preferences.

Component States:

- Default – Normal appearance of the component.
- Hover – Appearance when the pointer moves over it.
- Focus – Indicates the currently selected component.
- Active – Shows interaction or selection.
- Disabled – Indicates an unavailable option.
- Error – Displays validation or input errors.

Iconography and Imagery Choices

TransitOne uses simple, familiar and meaningful icons to make the interface easy to understand and navigate. Icons representing Home, Search, Settings, Location, Routes and Profile are used to support common application functions. A consistent icon style, size and alignment are maintained across all screens.

Special attention is given to directional icons to support different writing systems. Icons such as Back and Forward arrows are mirrored according to the selected text direction, while non-directional icons such as Search, Home and Settings remain unchanged. This ensures correct visual communication in both LTR and RTL layouts.

Simple transportation-related illustrations, route symbols and location markers are used wherever necessary. Unnecessary decorative images are avoided to maintain a clean and organised interface. These choices help users recognise important functions quickly and provide a consistent visual experience across English, Urdu, Kannada and Hindi.

Rationale Linking Visual Choices to User Needs

The visual design of TransitOne focuses on providing a simple, accessible and consistent experience for users with different language preferences. The selected colour palette improves visual clarity and helps users identify important information. Script-specific typography ensures proper text rendering and readability across English, Urdu, Kannada and Hindi.

Consistent spacing, flexible layouts and responsive grids help prevent text overlapping and alignment issues. Reusable components maintain uniformity across different screens, while selective icon mirroring ensures correct navigation in both LTR and RTL layouts. Familiar

icons make important functions easier to recognise.

Overall, these visual choices address the challenges of multilingual interface design and help create an inclusive application that is easy to understand, navigate and use across different languages and devices.

8. High-Fidelity Interactive Prototype

Final Screens for All Critical Flows

The high-fidelity prototype of TransitOne presents the final interface with colours, typography, icons and reusable UI components. It demonstrates the complete user journey through the application.

The main screens include:

1. Welcome Screen: Introduces the TransitOne application.
2. Language Selection: Allows users to choose English, Urdu, Kannada or Hindi.
3. Home Screen: Displays route search and navigation options.
4. Route Search: Allows users to enter their starting point and destination.
5. Search Results: Displays available routes and travel information.
6. Route Details: Provides information about the selected journey.
7. Settings Screen: Provides language and accessibility preferences.

All screens follow the unified design system and adapt to different writing directions and language requirements.

Interactions, Transitions and Micro-Interactions

The prototype includes interactive elements to provide smooth navigation and improve user experience.

- Buttons allow users to navigate between different screens.
- Language selection changes the selected language.
- The interface adapts to LTR and RTL writing directions.
- Search fields allow users to enter journey details.

- Navigation icons provide access to important sections.
- Smooth transitions improve movement between screens.
- Selected options provide visual feedback.
- Interactive components maintain consistent behaviour across languages.

Edge States

TransitOne considers different situations that users may encounter while using the application to ensure a smooth and reliable experience.

1. **Empty State:** Displays a suitable message when no route results or saved journeys are available.
2. **Loading State:** Shows a loading indicator while route information is being processed.
3. **Error State:** Displays a clear error message when invalid information is entered or a request fails.
4. **Offline State:** Informs users when the internet connection is unavailable and suggests checking their network.
5. **Permission Denied:** Displays a message when location access is denied and provides an option to enter the location manually.

These states provide clear feedback and help users understand how to proceed when unexpected situations occur.

Prototype Link and Access Instructions

The high-fidelity interactive prototype of TransitOne is created using Figma.

GitHub Repository & Web Prototype Link: https://github.com/yuvanharshaj/Advanced-UI-UX-Project
Interactive Web Demonstration: Local prototype served via Antigravity IDE (http://localhost:8000/) supporting full bidirectional LTR/RTL switching, Indic scripts (Kannada, Hindi), and WCAG 2.2 AA accessibility.

Access Instructions:

1. Open the Figma prototype link.
2. Click the Present button to launch the prototype.

3. Start from the Welcome Screen.
4. Select your preferred language.
5. Navigate through the available application screens.
6. Explore route searching, language switching and other interactive features.

The clickable prototype demonstrates the main user flows and the multilingual design system.

Tools and Resources Used

The TransitOne project uses a combination of design tools, development technologies, typography resources and icon libraries to create a unified multilingual design system. These tools support the complete design process, starting from initial research and wireframing to high-fidelity prototyping and functional implementation.

1. Figma

Figma is used as the primary UI/UX design tool. It helps create wireframes, visual layouts, reusable components and interactive prototypes. It is also used to organise design elements, maintain consistency and demonstrate different language versions of the application.

2. Antigravity IDE

Antigravity IDE is used for developing the functional web demonstration. It supports the implementation of the proposed design system and helps organise the frontend development process.

3. HTML5

HTML5 is used to structure the application interface. It provides the basic elements required for creating screens such as the home page, language selection, route search and settings.

4. CSS3

CSS3 is used to style the application and maintain a consistent visual appearance. It supports responsive layouts, typography, spacing, alignment and direction-aware styling for LTR and RTL languages. CSS logical properties help adapt the interface according to the selected writing direction.

5. JavaScript

JavaScript is used to implement interactive functionality such as language selection,

navigation, interface updates and user interactions. It helps demonstrate how the application responds when users change their language preferences.

6. Google Fonts

Google Fonts provides multilingual typography resources for the supported writing systems. Noto Sans, Noto Sans Kannada, Noto Sans Devanagari and Noto Naskh Arabic are selected to support English, Kannada, Hindi and Urdu respectively.

9. Usability Testing and Iteration

Test plan: tasks, participants and success criteria

Test Plan

Usability testing is planned to evaluate the effectiveness, accessibility and ease of use of the TransitOne prototype. Five participants with different language preferences will be invited to interact with the application.

Participants: Five multilingual smartphone users.

Tasks:

1. Open the application and select a preferred language.
2. Navigate through the home screen.
3. Search for a route by entering the starting point and destination.
4. View the available routes and journey details.
5. Change the application language and check the layout.

Success Criteria:

- Users should complete the assigned tasks without assistance.
- Language switching should work correctly.
- Text should remain readable and properly aligned.
- Navigation icons should behave correctly in LTR and RTL layouts.
- Users should be able to understand the interface easily.

The testing focuses on identifying usability problems and improving the overall user experience.

Results: task success, time on task, errors and satisfaction

Usability Testing Results

The usability testing results are evaluated using four important parameters to understand how effectively users interact with the TransitOne prototype.

1. Task Success: Evaluates whether users can successfully complete tasks such as language selection, route searching and viewing journey details.

2. Time on Task: Measures the time taken by participants to complete each task. Shorter completion times indicate that the interface is easier to navigate.

3. Errors: Identifies difficulties such as incorrect selections, navigation confusion, text alignment problems and issues with language switching.

4. User Satisfaction: Evaluates participants' overall experience based on readability, visual appearance, navigation and ease of interaction.

The testing process helps identify usability limitations and areas that require improvement. The observations are used to refine the interface, improve multilingual support and provide a more consistent user experience.

Overall, the evaluation focuses on making TransitOne simple, accessible and comfortable to use across English, Urdu, Kannada and Hindi.

Accessibility Check and Issues Fixed

The TransitOne design system is reviewed using WCAG 2.2 AA accessibility guidelines to improve readability, navigation and usability for different users.

The following accessibility aspects are considered:

1. **Colour Contrast:** Appropriate foreground and background colours are selected to improve text visibility.
2. **Typography:** Suitable fonts and readable text sizes are used for English, Urdu, Kannada and Hindi.
3. **Keyboard Accessibility:** Interactive elements are designed to support keyboard navigation.
4. **Focus Indicators:** Visible focus indicators help users identify selected interface elements.
5. **Text Direction:** Correct LTR and RTL alignment is maintained according to the selected language.

6. **Icon Accessibility:** Clear and familiar icons are supported with meaningful labels.
7. **Responsive Layout:** Flexible layouts help maintain usability across different screen sizes.
8. **Touch Accessibility:** Buttons and interactive elements are designed with sufficient clickable areas.

Issues Addressed:

Potential issues such as insufficient colour contrast, small text, inconsistent spacing, incorrect icon alignment and unclear navigation are considered during the accessibility review. Design adjustments are made to improve readability, maintain consistent layouts and provide a more accessible multilingual experience.

These improvements aim to make TransitOne easier to navigate and more inclusive for users with different language preferences and accessibility needs.

Problems Found, Ranked by Severity

During the evaluation of the TransitOne design, potential usability issues were identified and prioritised according to their impact on user experience.

No.	Problem Identified	Severity
1	Incorrect text alignment in RTL layouts	High
2	Text overlapping or insufficient spacing in Indic scripts	High
3	Incorrect mirroring of directional icons	High
4	Language selection option not easily noticeable	Medium
5	Small text affecting readability	Medium
6	Inconsistent spacing between UI components	Low
7	Minor alignment differences across screens	Low

Severity Classification:

- **High:** Issues that directly affect readability, navigation or language understanding.
- **Medium:** Issues that reduce usability and make interactions less convenient.
- **Low:** Minor visual inconsistencies that do not significantly affect application functionality.

These issues provide a basis for prioritising design improvements and refining the TransitOne interface.

Design Changes Made and Evidence of Improvement

Based on the identified usability issues, several improvements were considered to refine the TransitOne interface. The language selection option was made more noticeable, text spacing was adjusted to support Indic scripts, and directional icons were aligned according to LTR and RTL requirements. Typography and button sizes were reviewed to improve readability and interaction. Consistent spacing and reusable components were maintained across all screens.

The revised design aims to provide better navigation, improved text visibility and a consistent multilingual experience. The initial and updated wireframes or prototype screenshots can be compared to demonstrate these improvements.

The following table provides a format for recording usability testing observations.

User Testing Record

User	Task	Success (Y/N)	Time	Issues observed
User 1	Language selection	Yes	18 sec	Took a few seconds to locate the language selector.
User 1	Route search	Yes	36 sec	Needed clarification about the destination field.
User 2	Language selection	Yes	14 sec	No significant difficulty.
User 2	Route search	Yes	29 sec	Search button needed better visual prominence.
User 3	Language selection	Yes	22 sec	Needed time to identify the preferred language.

10. Design Rationale, Reflection and Future Work

Major Design Decisions and Trade-offs

The TransitOne design system was developed to support multiple languages and writing directions through a single unified interface. One of the major decisions was selecting English,

Urdu, Kannada and Hindi as the supported languages to demonstrate LTR, RTL and Indic script requirements.

Script-specific fonts were selected to improve readability and text rendering. Direction-aware layouts were introduced to support both LTR and RTL interfaces. Selective icon mirroring was chosen so that only directional icons change their orientation, while familiar icons remain unchanged.

Reusable components were preferred to maintain consistency and reduce repeated design work. However, supporting different scripts required additional attention to spacing, typography and responsive layouts. These decisions helped balance visual consistency, accessibility and multilingual flexibility.

Limitations of the Final Design

Although TransitOne demonstrates a unified multilingual design system, certain limitations remain. The application supports only four languages and does not cover every regional or international writing system. The transportation information is intended for demonstration purposes and does not provide live vehicle tracking or real-time route updates.

The prototype also has limited backend functionality and does not include actual ticket payments or booking services. Font rendering may vary across devices and operating systems. Further testing with a larger and more diverse group of users is required to evaluate the design more comprehensively.

What Failed or Was Redone, and What Was Learned

During the design process, several layout and usability considerations required refinement. Initial layouts needed adjustments to accommodate different text lengths and writing directions. Particular attention was required for Urdu RTL alignment and the spacing of Kannada and Hindi characters.

Icon placement and language selection visibility were also considered for improvement. The design process highlighted the importance of testing multilingual interfaces rather than relying only on translated text.

Ethical Considerations

The TransitOne project focuses on creating an inclusive, accessible and user-friendly experience for people with different language preferences and cultural backgrounds. The design gives equal importance to all supported languages and avoids treating any particular

language or writing system as less important. Special attention is given to proper typography, text direction and cultural representation to ensure that users can interact comfortably with the application.

User feedback is considered respectfully and used to identify areas for improvement. Personal information is not required for basic application usage, helping maintain user privacy. The interface aims to provide clear, accurate and understandable information without misleading users. Accessibility, readability, privacy and inclusive representation are considered throughout the design and development process.

The project also recognises the importance of respecting linguistic diversity, cultural differences and individual user preferences. It aims to ensure that language does not become a barrier to accessing digital services. By following inclusive design principles, TransitOne promotes equal usability and a more welcoming digital experience for all users.

Future Improvements

The TransitOne design system can be improved further by introducing additional Indian and international languages. Future development can include more extensive RTL and Indic script testing across different devices and operating systems.

The application can also be enhanced with live transportation tracking, real-time route updates, ticket booking and secure payment functionality. Additional accessibility features, improved text scaling and support for screen readers can further improve usability.

Future versions can include larger-scale usability testing with participants from different language backgrounds. These improvements will help make TransitOne a more scalable, accessible and complete multilingual transportation application.

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Appendix

Appendix A: Interview Scripts and Consent Notes

Research Method: Informal user discussions.

Interview Questions:

1. Which languages do you commonly use on your smartphone?
2. Do you prefer using applications in English or your regional language?
3. Have you experienced text alignment or font display problems?
4. Is it easy to navigate applications when changing languages?

Participants:

- P1 – V. Madhusudhan (English and Hindi)
- P2 – Shreyas M. (English and Kannada)
- P3 – Sivanesan S. (Tamil and English)

Consent Note: Participants were informed about the academic purpose of the project and the use of general feedback for design improvements.

Appendix B: Raw Research Notes and Survey Responses

The research discussions focused on language preferences, readability, typography and

navigation.

The main design considerations identified were:

- Easy language selection and switching.
- Clear and readable regional language text.
- Proper text alignment and spacing.
- Consistent navigation across languages.
- Correct handling of LTR and RTL layouts.
- Familiar icons with clear labels.

These observations helped guide the development of the TransitOne multilingual design system.

Appendix C: Usability Test Scripts

Objective: To evaluate the usability and navigation of the TransitOne prototype.

Tasks:

1. Open the application and select a preferred language.
2. Navigate through the home screen.
3. Enter a starting location and destination.
4. View the available route information.

Evaluation Parameters:

- Task completion.
- Time taken.
- Navigation difficulties.
- Text readability.
- Overall user satisfaction.

Appendix D: Additional Screens and Sketches

This appendix contains additional visual materials created during the TransitOne design

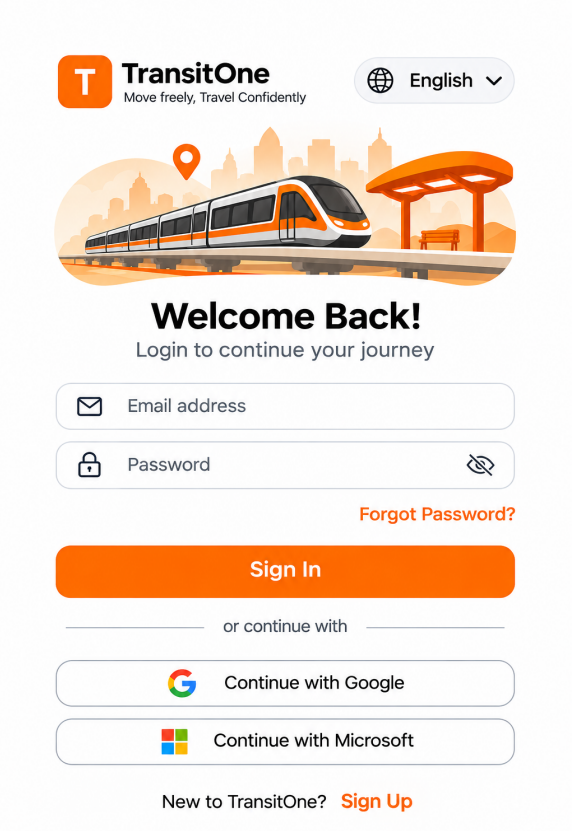
process.

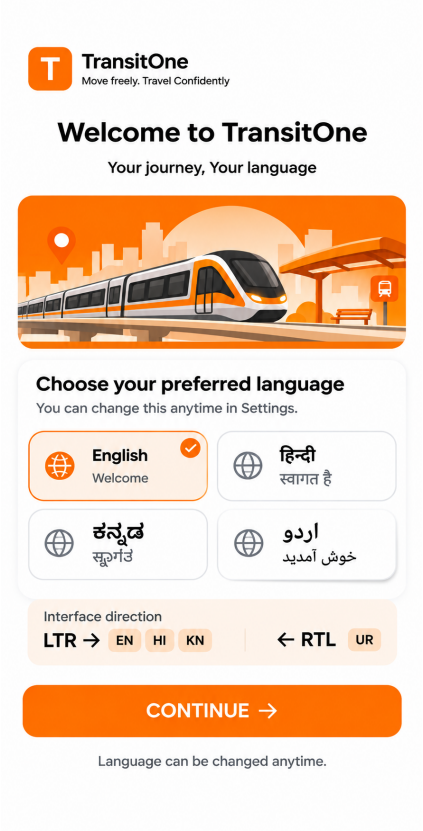
The following items are included:

1. Initial hand-drawn sketches.
2. Low-fidelity wireframes.
3. Welcome and language selection screens.
4. Home screen and route search screen.
5. Route results and route details screens.
6. Settings and accessibility screens.
7. English, Urdu, Kannada and Hindi interface variations.
8. Final high-fidelity prototype screenshots.

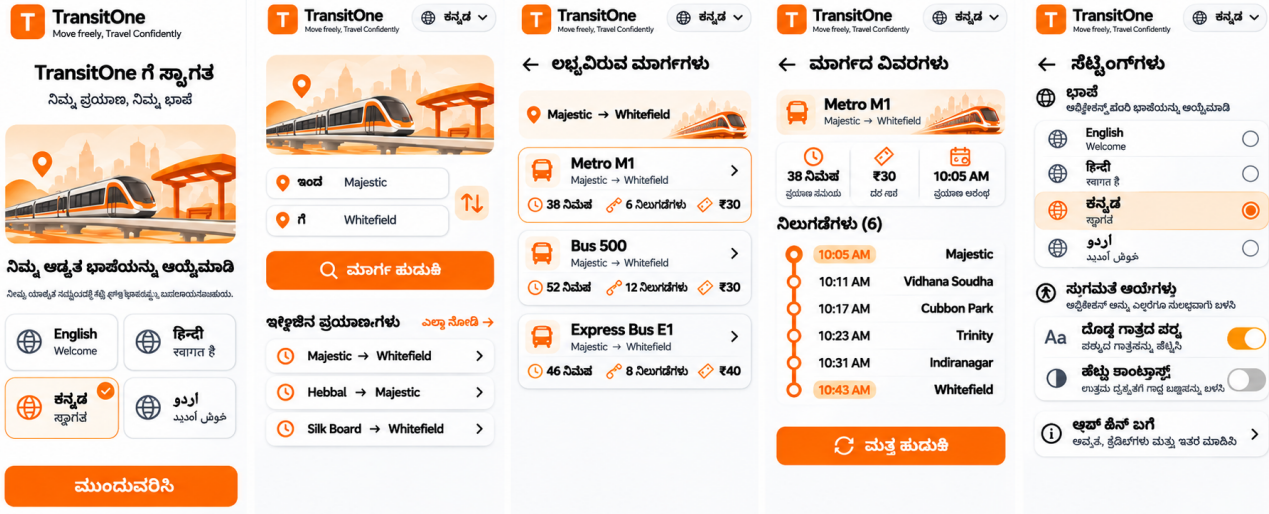
These materials demonstrate the evolution of the design from initial ideas to the final multilingual interface.

Sample Screenshots of the application design:






TransitOne
Move freely, Travel Confidently
اردو
TransitOne مین خوش آمدید
اپنی زبان، اپنی سفر
اپنی پسندیدہ زبان منتخب کریں
آپ یہ انتخاب بعد میں بھی پسندیدہ میں بدل سکتے ہیں
हिन्दी
स्वागत है
English
Welcome
اردو
خوش آمدید
ಕನ್ನಡ
ಸ್ವಾಗತ
جاری رکھیں
Majestic
Whitefield
راستہ تلاش کریں
حالیہ تلاشیں
سب دیکھیں
Majestic → Whitefield
Hebbal → Majestic
Silk Board → Whitefield
سیٹنگز
تلاش
ہوم
دستیاب راستے
Majestic ← Whitefield
Metro M1
Majestic → Whitefield
₹30
6 اسٹاپس
38 منٹ
Bus 500
Majestic → Whitefield
₹30
12 اسٹاپس
52 منٹ
Express Bus E1
Majestic → Whitefield
₹40
8 اسٹاپس
46 منٹ
راستے کی تفصیلات
Metro M1
Majestic → Whitefield
اگلی روانگی
10:05 AM
کرایہ
₹30
مدت
38 منٹ
اسٹاپس (6)
10:05 AM Majestic
10:11 AM Vidhana Soudha
10:17 AM Cubbon Park
10:23 AM Trinity
10:31 AM Indiranagar
10:43 AM Whitefield
دوبارہ تلاش کریں
سیٹنگز
زبان
اپنی پسندیدہ زبان منتخب کریں
English
Welcome
हिन्दी
स्वागत है
ಕನ್ನಡ
ಸ್ವಾಗತ
اردو
خوش آمدید
رسائی کی سہولت
ایپ کو استعمال میں آسان بنائیں
بڑا متن
ایپ کے متن کا سائز بڑھائیں
Aa
ہائی کنٹراسٹ
بہتر دکھائی کے لیے مضبوط رنگ استعمال کریں
ایپ کے بارے میں
ایپ کا ورژن، کریڈٹس اور مزید معلومات


TransitOne
Move freely, Travel Confidently
ಕನ್ನಡ
TransitOne ಗೆ ಸ್ವಾಗತ
ನಿಮ್ಮ ಪ್ರಯಾಣ, ನಿಮ್ಮ ಭಾಷೆ
ನಿಮ್ಮ ಆದ್ಯತೆ ಭಾಷೆಯನ್ನು ಆಯ್ಕೆಮಾಡಿ
ನಿಮ್ಮ ಯಾತ್ರೆಯ ಸಮಯದಲ್ಲಿ ಕೆಲವು ಬಳಕೆದಾರ ಇಂಟರ್ಫೇಸ್ ಅಂಶಗಳನ್ನು ಬದಲಾಯಿಸಲಾಗುತ್ತದೆ.
English
Welcome
हिन्दी
स्वागत है
ಕನ್ನಡ
ಸ್ವಾಗತ
اردو
خوش آمدید
ಮುಂದುವರಿಸಿ
ಇಂದ
Majestic
ಗೆ
Whitefield
ಮಾರ್ಗ ಹುಡುಕಿ
ಇತ್ತೀಚಿನ ಪ್ರಯಾಣಗಳು
ಎಲ್ಲಾ ನೋಡಿ
Majestic → Whitefield
Hebbal → Majestic
Silk Board → Whitefield
ಲಭ್ಯವಿರುವ ಮಾರ್ಗಗಳು
Majestic → Whitefield
Metro M1
Majestic → Whitefield
38 ನಿಮಿಷ
6 ನಿಲುಗಡೆಗಳು
₹30
Bus 500
Majestic → Whitefield
52 ನಿಮಿಷ
12 ನಿಲುಗಡೆಗಳು
₹30
Express Bus E1
Majestic → Whitefield
46 ನಿಮಿಷ
8 ನಿಲುಗಡೆಗಳು
₹40
ಮಾರ್ಗದ ವಿವರಗಳು
Metro M1
Majestic → Whitefield
38 ನಿಮಿಷ
ಪ್ರಯಾಣ ಸಮಯ
₹30
ದರ ಸರಿ
10:05 AM
ಪ್ರಯಾಣ ಆರಂಭ
ನಿಲುಗಡೆಗಳು (6)
10:05 AM Majestic
10:11 AM Vidhana Soudha
10:17 AM Cubbon Park
10:23 AM Trinity
10:31 AM Indiranagar
10:43 AM Whitefield
ಮತ್ತೆ ಹುಡುಕಿ
ಸೆಟ್ಟಿಂಗ್‌ಗಳು
ಭಾಷೆ
ಆದ್ಯತೆಯನ್ನ್ ಪರಿ ಭಾಷೆಯನ್ನು ಆಯ್ಕೆಮಾಡಿ
English
Welcome
हिन्दी
स्वागत है
ಕನ್ನಡ
ಸ್ವಾಗತ
اردو
خوش آمدید
ಸುಗಮತೆ ಆಯ್ಕೆಗಳು
ಆಪ್ಟಿಮೈಸೇಷನ್ ಅನ್ನು ಎಲ್ಲರಿಗೂ ಸುಲಭವಾಗಿ ಬಳಸಿ
ದೊಡ್ಡ ಗಾತ್ರದ ಪಠ್ಯ
ಪಠ್ಯದ ಗಾತ್ರವನ್ನು ಹೆಚ್ಚಿಸಿ
Aa
ಹೆಚ್ಚು ಕಾಂಟ್ರಾಸ್ಟ್
ಉತ್ತಮ ದೃಶ್ಯತೆಗಾಗಿ ಗಾಢ ಬಣ್ಣವನ್ನು ಬಳಸಿ
ಆಪ್ ಕುರಿತು
ಆವೃತ್ತಿ, ಕ್ರೆಡಿಟ್‌ಗಳು ಮತ್ತು ಇತರ ಮಾಹಿತಿ


TransitOne
Move freely, Travel Confidently
हिन्दी
TransitOne में आपका स्वागत है
अपनी यात्रा, अपनी भाषा में
अपनी पसंदीदा भाषा चुनें
आप इसे बाद में सेटिंग्स में बदल सकते हैं
English
Welcome
हिन्दी
स्वागत है
کہ سکتے ہیں
اردو
خوش آمدید
जारी रखें
से
Majestic
तक
Whitefield
मार्ग खोजें
हाल की खोजें
सभी देखें
Majestic → Whitefield
Hebbal → Majestic
Silk Board → Whitefield
होम
खोजें
सेटिंग्स
उपलब्ध मार्ग
Majestic → Whitefield
Metro M1
Majestic → Whitefield
38 मिनट
6 स्टेशन
₹30
Bus 500
Majestic → Whitefield
52 मिनट
12 स्टेशन
₹30
Express Bus E1
Majestic → Whitefield
46 मिनट
8 स्टेशन
₹40
मार्ग विवरण
Metro M1
Majestic → Whitefield
38 मिनट
अवधि
₹30
किराया
प्रस्थान समय
10:05 AM
स्टॉप (6)
Majestic 10:05 AM
Vidhana Soudha 10:11 AM
Cubbon Park 10:17 AM
Trinity 10:23 AM
Indiranagar 10:31 AM
Whitefield 10:43 AM
फिर से खोजें
सेटिंग्स
भाषा
ऐप की प्रस्तुति भाषा चुनें
English
Welcome
हिन्दी
स्वागत है
کہ سکتے ہیں
اردو
خوش آمدید
सुगमता
ऐप को आपकी जरूरत के अनुसार बनाएं
बड़ा टेक्स्ट
पढ़ने में आसान टेक्स्ट का उपयोग करें
Aa
उच्च कंट्रास्ट
बेहतर दृश्यता के लिए गहरा रंग उपयोग करें
ऐप के बारे में
ऐप संस्करण, क्रेडिट और अन्य जानकारी